

Toronto Rush Hour Traffic Analysis*

INF2167 Project - Group 7

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*Code and data are available at: https://github.com/srg-uoft/INF2167_Group7_Project.

#Introduction Urban mobility in the Greater Toronto Area (GTA) is shaped by dense traffic volumes, multimodal travel patterns, and complex interactions among different groups (drivers, cyclists, pedestrians, and people taking transits). Understanding conditions and causes of collisions are essential for improving road safety by designing targeted interventions, and aligning municipal planning with evidence based risk patterns. This project adopts a data first approach in exploring a up to date open dataset published by the Toronto Police Service (TPS). The dataset enables a investigation of collision patterns across time, location, and mode of travel, via its detailed records in large scale. In this way, we can examine whether collision trends in Toronto align with broader research on high risk travel periods including holidays, along with neighborhoods and modes of transportation with highest safety risks.

1 Research Questions(s) & Hypothesis

1.1 Questions

What are the most accident-prone locations and times for travellers in the GTA, and how does that vary depending on mode of transportation? The processed dataset allows us to compare risk profiles across modes, thanks to a indicator for involvements of different modes of travel. *Does the GTA's data align with pre-existing research on riskiest travel times?* Prior research frequently identifies holidays as high risk periods for collisions. By analyzing temporal patterns in the TPS dataset, we can test whether Toronto follows these broader trends or shows unique local patterns. This comparison helps contextualize Toronto's traffic safety landscape within the wider literature on collision risk.

1.2 Hypotheses

Before examining the dataset, we have the hypothesis that GTA traffic risk increases during well known high risk periods, such as New Year's Eve and other major holidays, as well as during late night hours and winter months.

2 Relevant Literature

Scholars consistently conclude that road traffic collisions cluster in both location and time, thus we are inspired on discovering when and where risk is highest in the GTA. Vaz, Tehranchi, and Cusimano (2017) conduct a spatial analysis of road traffic injuries in the Greater Toronto Area using Moran's I and Getis Ord Gi. The article concludes that collisions form statistically significant clusters rather than occurring randomly. Their results show that "Peel and Durham have the highest collision rate", which reminds us of the importance to identify hotspots of collisions in different neiboughoods. This directly provides us preview on Toronto collisions

by communities. Temporal patterns in collision risk are also documented already by scholars. Anowar, Yasmin, and Tay (2013) compare crashes during statutory holidays and regular weekends in Alberta. They find that severe collision occurrences are less than estimated. Certain days (especially major holidays) carry elevated collision risk, which inspires our interest in comparing GTA collision patterns to commonly recognized high risk days.

3 Dataset & Methods

3.1 Dataset

3.1.1 Source

The data used in this analysis is the Traffic Collisions Open Data (ASR-T-TBL-001) dataset, obtained from (Toronto Police Service 2023) and licensed under the Open Government Licence – Ontario. The Toronto Police Service (TPS) maintains a Public Safety Data Portal, where they regularly publish and update datasets regarding crime statistics in the Greater Toronto Area, including the Traffic Collisions Open Data dataset.

3.1.2 Structure

TPS defines a collision for the purposes of this dataset as “the contact resulting from the motion of a motor vehicle or streetcar or its load, which produces property damage, injury or death” (Toronto Police Service 2023). To that end, each row/observation in the dataset represents a single collision, and the columns/variables describe the collision in three ways: 1) spatial/temporal information about where and when the collision occurred, 2) information about whether there were injuries, deaths, or property damage, and whether parties failed to remain at the scene, and 3) which types of vehicles were involved in the collision.

3.1.3 Key Variables

To answer the question of where the highest risk of traffic collision for various modes of transportation is in the Greater Toronto Area, our primary variables of interest are the spatial and temporal variables (to determine where and when the most accidents occur), and the information about involved vehicles (to determine any diverging trends between modes of transportation).

3.2 Methods

Methods we plan to employ include line charts for trends in accidents over time, and bar charts showing accident frequencies for each type of vehicle in each month. We also plan on utilizing summary statistic tables, particularly to show the most dangerous areas of Toronto for individual types of vehicles. This will likely involve the use of data pivoting to turn separate transportation columns into individual rows, as well as grouping and faceting to show differences in collision frequency.

4 Exploratory Data Analysis

4.1 Overview

Initial exploratory data analysis was done using statistical programming language (R Core Team 2023) and the Tidyverse library (Wickham et al. 2019), with the goal of uncovering high-level trends and possible variables of interest for further analysis. Filepaths were made relative using the Here package (Müller 2025) to improve reproducibility, and color palettes are courtesy of (Hvitfeldt 2021). Key discoveries which we hope will aid in answering our research question in future reports are elaborated on below.

4.2 Separated Temporal Variables

As mentioned in (Alexander 2023), dates can be a tricky form of data to work with, as their class and formatting can very often not work with common data analysis methods, and they can take significant cleaning to make useful. However, this dataset not only has the data represented in string format, it also has separate columns with the year and hour of the collision represented as doubles, and day of the week and month represented as character strings. Given our focus on finding the riskiest times for travellers in various transportation modes to be on the road in Toronto, the ability to count collisions for individual months or hours, as well as specific dates, will be helpful in drawing conclusions. A preliminary example is shown in (Figure 1), which shows which months have the most collisions reported.

4.3 Missing Spatial Data

An aspect of our research we may encounter more issues with is spatial data. The dataset includes a column for which Toronto neighbourhood each collision occurred in, as well as a crash latitude and longitude, which could be useful in determining where travellers face the highest risk. However, after visualizing where the most crashes are taking place, we see in (Table 1) that “NAS”, or crashes where no spatial data was recorded, is the most frequently occurring value in the Neighbourhood column, making up around 17% of the entries. We will

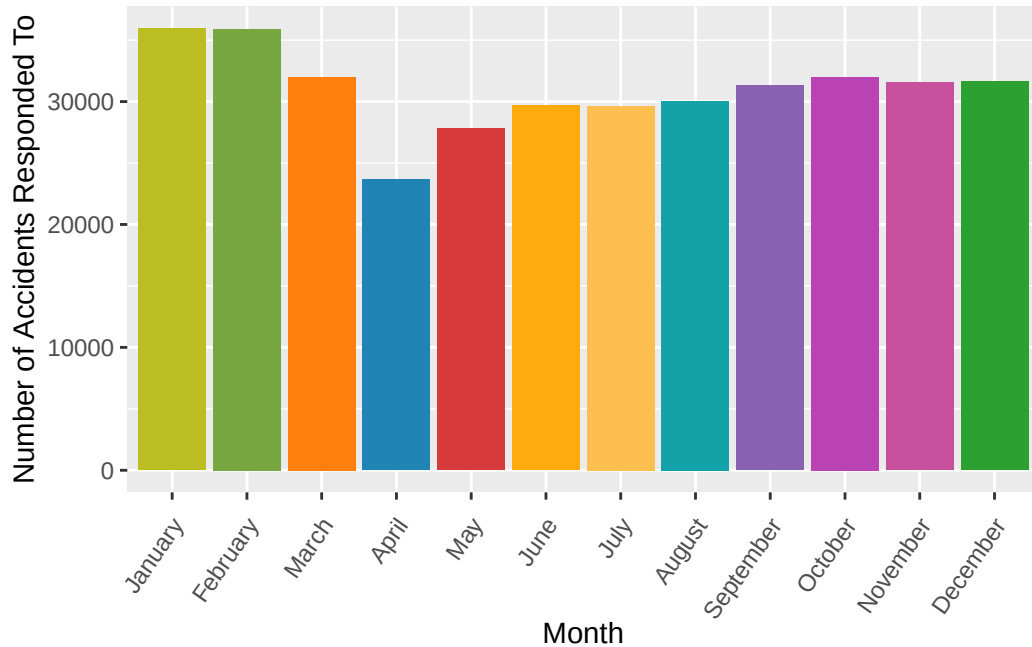


Figure 1: Accidents per Month from 2020-2026

have to determine doing forward whether these should be dropped, or whether our research question could be amended to omit spatial investigation.

4.4 Types of Vehicles Stored as Columns

A key part of our analysis is investigating how risk of collision for travellers changes with mode of transportation. To this end, the TPS dataset contains 5 columns, each of which provides a yes/no as to whether cars, public transit vehicles, motorcycles, bicycles, or pedestrians were included in the accident. While this is useful data to have, it is not tidy data, and we will need to pivot the dataframe so each row is an individual vehicle involved in an accident before we can determine trends by vehicle type. (Figure 2) shows a basic use case for this pivoted data, and giving us preliminary data about how frequently the various modes of transportation are involved in collisions.

5 Next Steps

Having outlined the research question of interest, target data set, and preliminary exploratory data analysis above, we will move into more detailed and thorough analysis of the dataset to identify any trends which could help address the primary research question. Conclusions and

Table 1

```
# A tibble: 159 x 3
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NEIGHBOURHOOD_158	Number_of_Collisions	Percent_of_Total
<chr>	<int>	<dbl>
1 NSA	62076	16.7
2 Wexford/Maryvale (119)	9853	2.65
3 West Humber-Clairville (1)	7494	2.02
4 York University Heights (27)	6089	1.64
5 Dorset Park (126)	5786	1.56
6 St Lawrence-East Bayfront-The Islands ~	5389	1.45
7 Milliken (130)	4988	1.34
8 Banbury-Don Mills (42)	4870	1.31
9 Etobicoke City Centre (159)	4775	1.29
10 Woburn North (142)	4746	1.28

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# i 149 more rows
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[1] 10

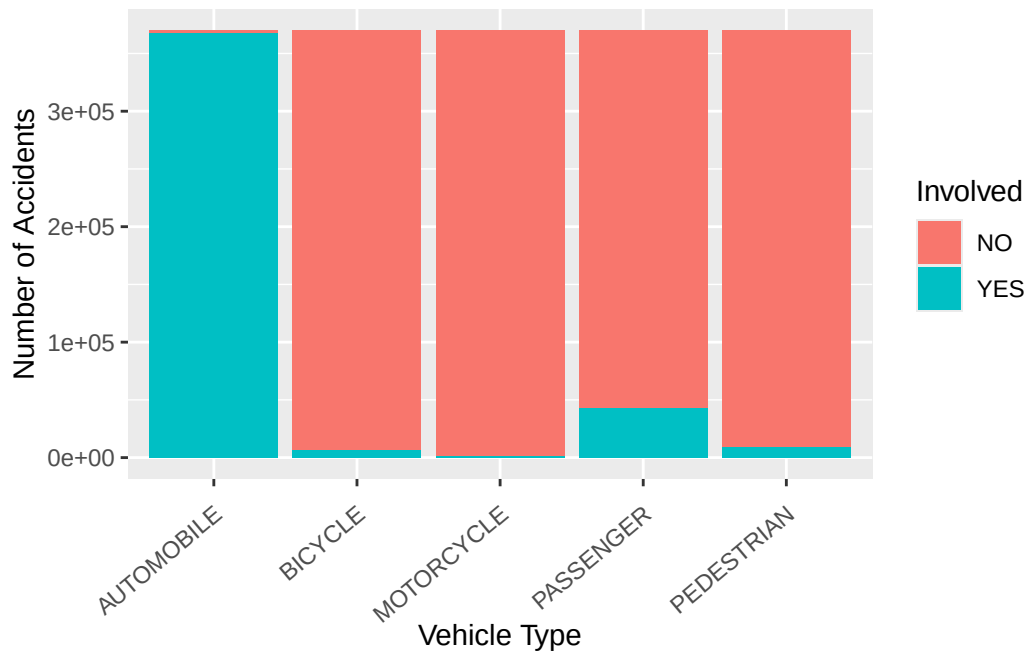


Figure 2: What Types of Vehicles Were Involved in Each Recorded Traffic Collision from 2020-2026

discussion, along with raw results and any models used in their generation, will be presented in a final report and oral presentation in August 2026.

6 References

- Alexander, Rohan. 2023. *Telling Stories with Data*. Chapman; Hall/CRC. <https://tellingstorieswithdata.com/>.
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- Wickham, Hadley, Mara Averick, Jennifer Bryan, et al. 2019. “Welcome to the Tidyverse.” *Journal of Open Source Software* 4 (43): 1686. <https://doi.org/10.21105/joss.01686>.